Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/26/2025
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
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City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	8
No. of PhD Candidates:	8	**No. of Master' Degrees:**	6
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 04/30/2025 per NSSC information (Ed., 12/3/24) NOTE: End date changed to 04/30/2024 per NSSC information (Ed., 11/6/23) NOTE: End date changed to 04/30/2023 per D. Stockton/GRC (Ed., 6/6/22) NOTE: End date changed to 06/11/2022 per NSSC information (Ed., 5/17/22) NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We completed Round 1 (normal flames) and Round 2 (inverse flames) of Flame Design testing aboard the International Space Station (ISS) and processed and analyzed the data. We performed extensive analyses of the data and conducted numerical modeling, primarily with Sphdif. We also conducted many experiments of laminar flames in normal gravity. Thirty-four journal papers and 67 conference papers were published. [Ed. Note: See Cumulative Bibliography.] This body of work was recognized with five awards. This project resulted in 16 theses and dissertations. Most of the associated students are now employed in academia, government or industry, and many are conducting research.
Bibliography Type:	Description: (Last Updated: 11/26/2025)